1. Carga de datos

In [2]:
```
import json
import pandas as pd
import matplotlib.pyplot as plt

with open("../../data/experiment_analysis.json") as f:
    data = json.load(f)

topics = data["topics"]

df = pd.DataFrame.from_dict(topics, orient="index")
df.reset_index(inplace=True)
df.rename(columns={"index": "topic"}, inplace=True)
```

2. Caracterización del tráfico

2.1. Ancho de banda

In [3]:
```
df["bw_percent"] = df["bandwidth_mb_s"] / df["bandwidth_mb_s"].sum() * 10
df.sort_values(by="bw_percent", ascending=False)[["topic", "bandwidth_mb_
```

Out[3]:

	topic	bandwidth_mb_s	bw_percent
36	/robot/front_rgbd_camera/depth/points	71.088803	70.868156
40	/robot/front_rgbd_camera/depth/image_raw	8.886586	8.859003
50	/robot/front_rgbd_camera/rgb/image_raw	6.665086	6.644399
77	/robot/robot_front_ptz_camera/image_raw	6.665071	6.644384
28	/robot/move_base/GlobalPlanner/potential	3.094075	3.084471
64	/gazebo/link_states	1.972124	1.966003
69	/gazebo/model_states	1.099804	1.096390
37	/robot/front_laser/scan	0.331207	0.330179
60	/robot/move_base/global_costmap/costmap_updates	0.069513	0.069297
1	/robot/map	0.068757	0.068544

In [4]:
```
df_sorted = df.sort_values(by="bw_percent", ascending=False)
df_sorted["cum"] = df_sorted["bw_percent"].cumsum()
n_topics = len(df_sorted)
top_95 = df_sorted[df_sorted["cum"] <= 95]

print("Top topics:", len(top_95))
print("Percentage:", len(top_95) / n_topics * 100)
```

```
Top topics: 4
Percentage: 4.761904761904762
```

2.2. Frecuencia

In [5]:
```
df.sort_values(by="frequency_hz", ascending=False)[["topic", "frequency_h
```

Out[5]:

	topic	frequency_hz
70	/clock	997.216015
64	/gazebo/link_states	997.069490
69	/gazebo/model_states	996.739807
8	/tf	136.928093
58	/robot/joint_states	100.131873
75	/robot/joint_pan_position_controller/state	100.021979
26	/robot/joint_tilt_position_controller/state	99.820506
78	/robot/imu/data_raw	98.684934
15	/robot/imu/rpy/filtered	98.684934
7	/robot/imu/data	98.684934

2.3. Tamaño

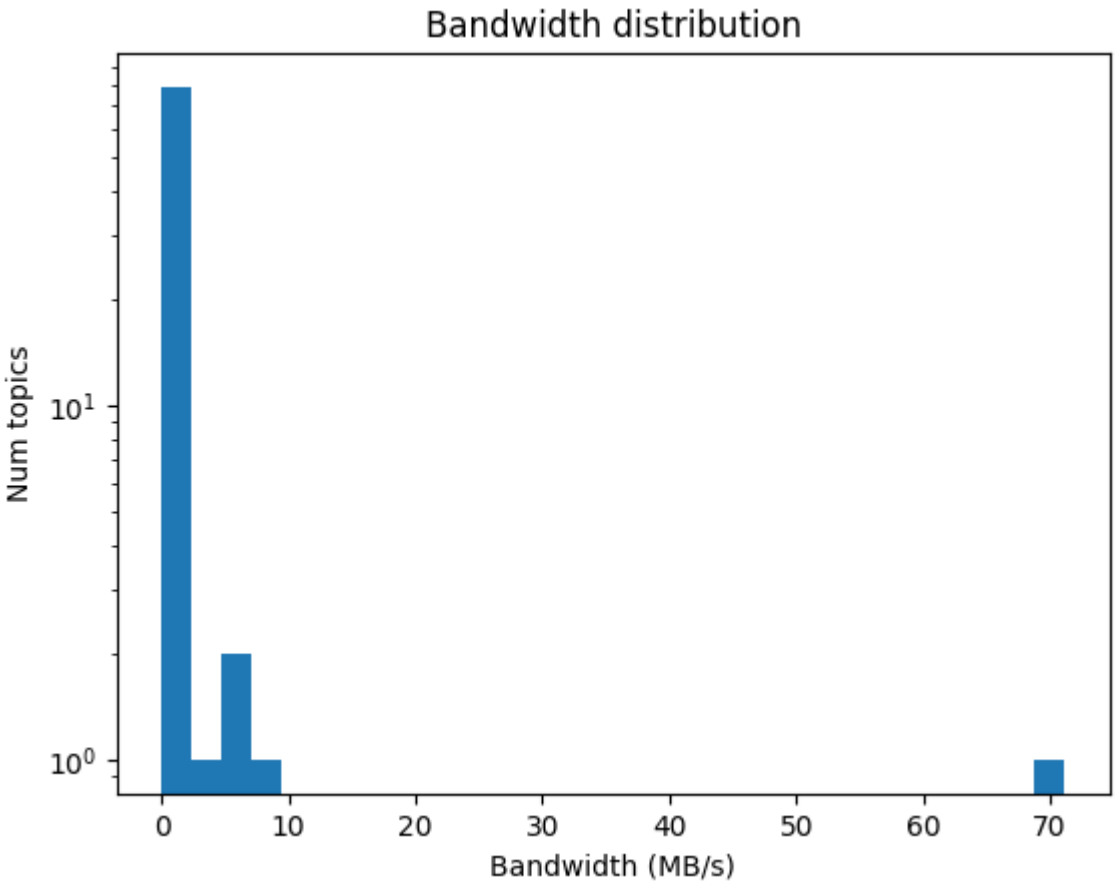
In [6]:
```
df.sort_values(by="avg_size_kb", ascending=False)[["topic","avg_size_kb"]
```

Out[6]:

	topic	avg_size_kb
36	/robot/front_rgbd_camera/depth/points	9600.139648
1	/robot/map	3844.102539
28	/robot/move_base/GlobalPlanner/potential	3844.102539
66	/robot/move_base/global_costmap/costmap	3844.102539
40	/robot/front_rgbd_camera/depth/image_raw	1200.083008
50	/robot/front_rgbd_camera/rgb/image_raw	900.082031
77	/robot/robot_front_ptz_camera/image_raw	900.080078
60	/robot/move_base/global_costmap/costmap_updates	82.687500
45	/robot/move_base/local_costmap/costmap	39.166016
63	/robot/particlecloud	31.696289

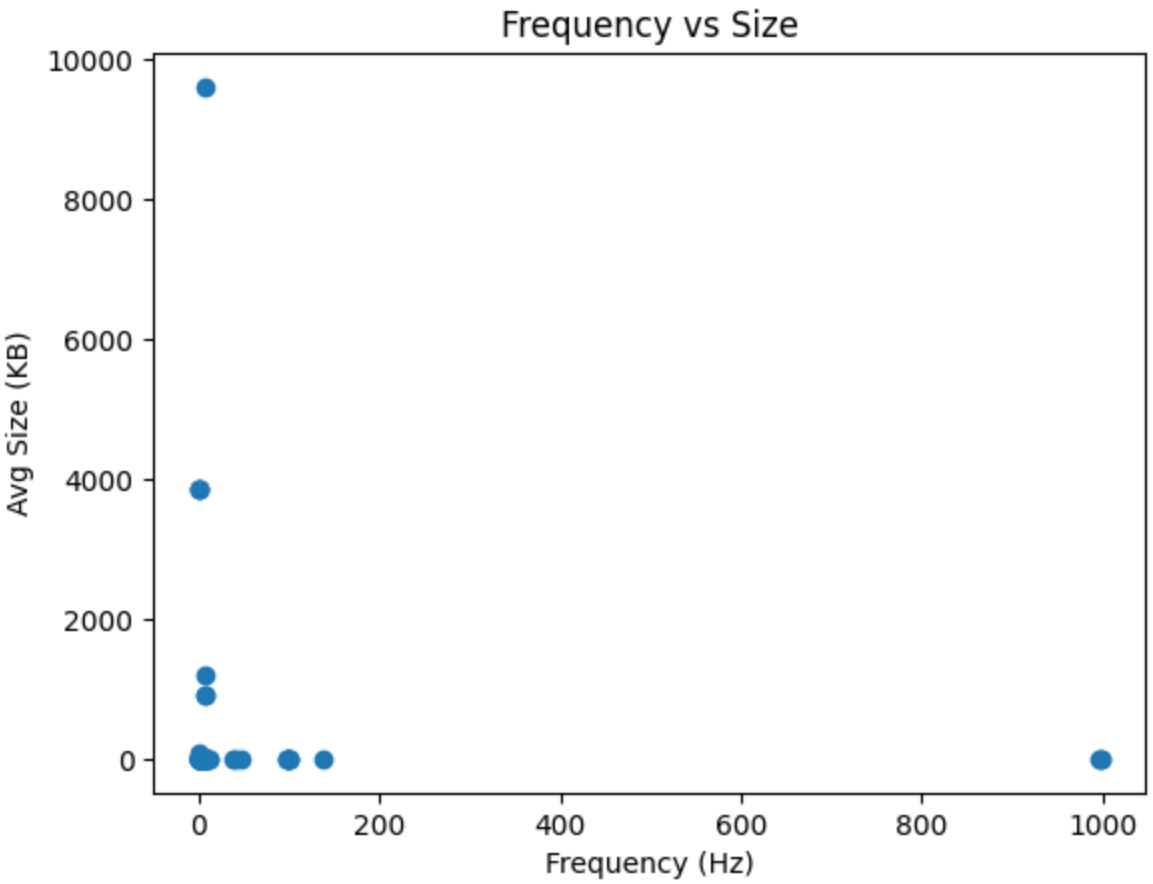
2.4. Distribución global

In [7]:
```
plt.hist(df["bandwidth_mb_s"], bins=30)
plt.xlabel("Bandwidth (MB/s)")
plt.ylabel("Num topics")
plt.title("Bandwidth distribution")
plt.yscale("log")
plt.show()
```



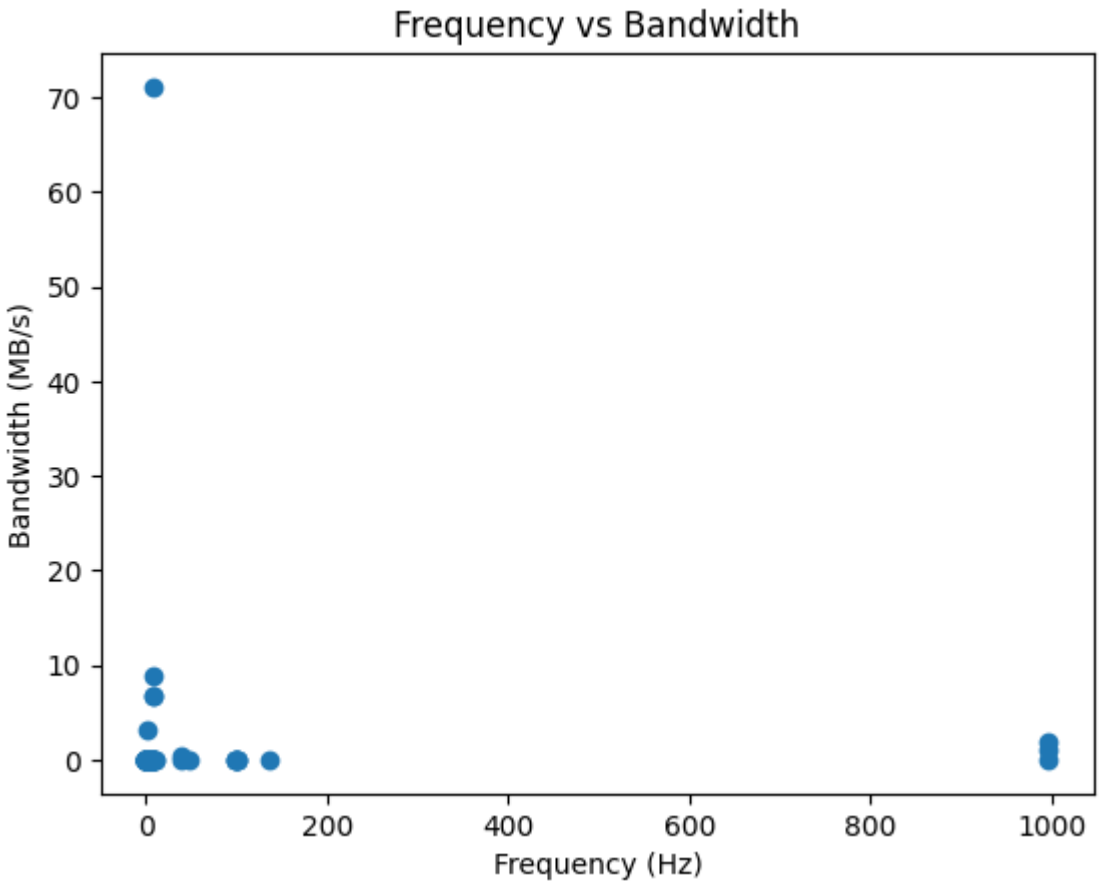
2.5. Relación frecuencia vs tamaño

In [8]:
```
plt.scatter(df["frequency_hz"], df["avg_size_kb"])
plt.xlabel("Frequency (Hz)")
plt.ylabel("Avg Size (KB)")
plt.title("Frequency vs Size")
plt.show()
```



2.6. Relación frecuencia vs bandwidth

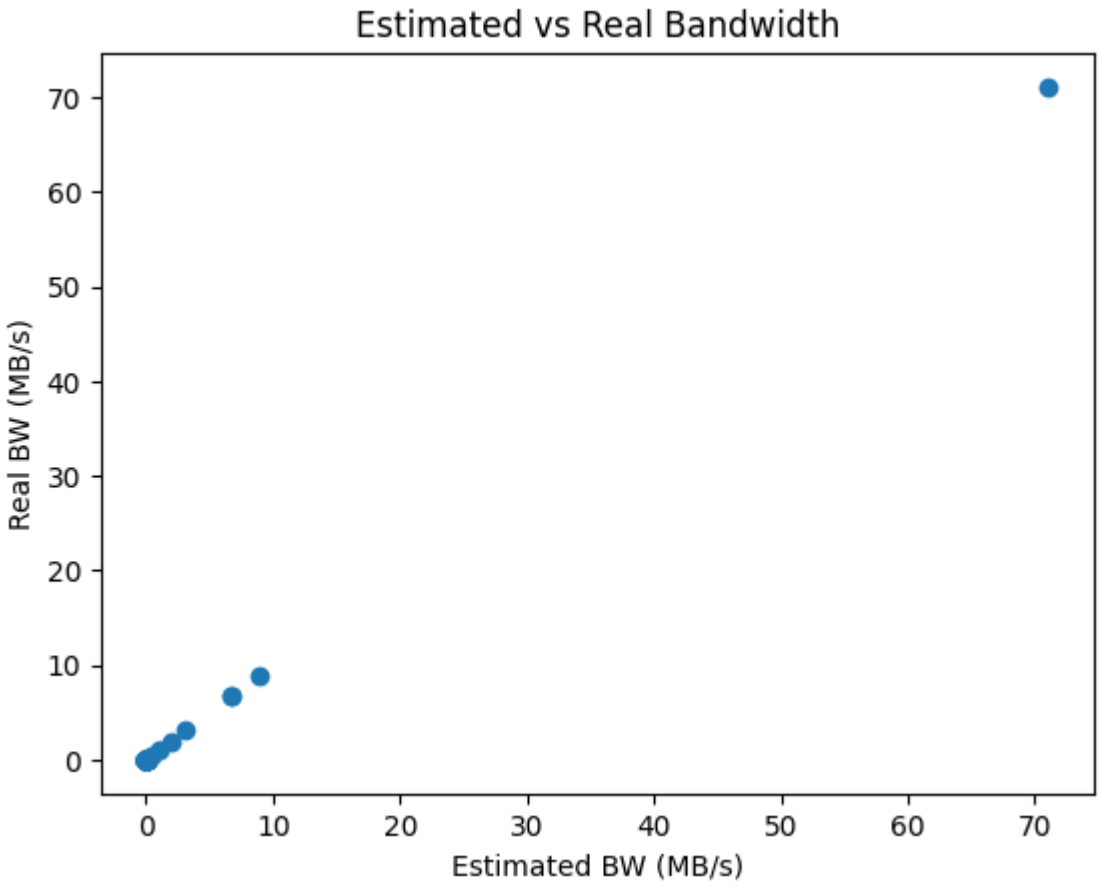
In [9]:
```
plt.scatter(df["frequency_hz"], df["bandwidth_mb_s"])
plt.xlabel("Frequency (Hz)")
plt.ylabel("Bandwidth (MB/s)")
plt.title("Frequency vs Bandwidth")
plt.show()
```



2.7. Validación BW ≈ frecuencia × tamaño

In [10]:
```
df["bw_estimated"] = df["frequency_hz"] * df["avg_size_kb"] / 1024

plt.scatter(df["bw_estimated"], df["bandwidth_mb_s"])
plt.xlabel("Estimated BW (MB/s)")
plt.ylabel("Real BW (MB/s)")
plt.title("Estimated vs Real Bandwidth")
plt.show()
```



In [11]:
```
from sklearn.metrics import r2_score

r2 = r2_score(df["bandwidth_mb_s"], df["bw_estimated"])
print("R2:", r2)
```

```
R2: 0.9999999999983807
```

2.8. Conclusiones

El tráfico del sistema presenta una distribución fuertemente concentrada, donde un subconjunto reducido de *topics* acumula la mayor parte del ancho de banda total. Empíricamente, aproximadamente el 95% del tráfico se concentra en torno al X% de los *topics*, confirmando un comportamiento tipo long-tail.

El análisis muestra que el ancho de banda está principalmente determinado por el tamaño de los mensajes y no por la frecuencia de publicación. Esto se valida mediante la fuerte correlación entre el ancho de banda real y el estimado (BW ≈ frecuencia × tamaño).

Los *topics* dominantes corresponden a sensores de percepción (cámaras RGB-D y *point clouds*), lo que indica que el sistema está fuertemente condicionado por la carga de datos sensoriales.

En contraste, la mayoría de los *topics* tienen un impacto marginal en el tráfico global.

3. Selección de flujo crítico

3.1. Filtrado de flujo crítico

In [12]:
```
BW_THRESHOLD = df["bandwidth_mb_s"].quantile(0.9)

critical_topics = df[df["bandwidth_mb_s"] > BW_THRESHOLD].copy().sort_val
```

3.2. Conclusiones

La selección basada en percentil (p90) permite aislar los *topics* que dominan el consumo de red. Estos *topics* representan los principales candidatos para optimización, ya que pequeñas mejoras en ellos tienen un impacto significativo en el sistema global.

Este filtrado reduce el espacio de análisis sin pérdida relevante de información, centrando el estudio en los flujos realmente críticos.

4. Analisis estructural

4.1 Carga grafo

In [13]:
```
with open("../../data/node_topic_graph.json") as f:
    graph = json.load(f)
```

4.2. Dependencias

In [14]:
```
rows = []

for _, row in critical_topics.iterrows():
    topic = row["topic"]
    pubs = graph.get(topic, {}).get("publishers", [])
    subs = graph.get(topic, {}).get("subscribers", [])

    rows.append({
        "topic": topic,
        "bw": row["bandwidth_mb_s"],
        "publishers": pubs,
        "subscribers": subs,
        "n_pub": len(pubs),
        "n_sub": len(subs)
    })

df_dep = pd.DataFrame(rows).sort_values(by="bw", ascending=False)
```

In [15]:
```
df_dep[df_dep["n_sub"] == 0]
```

Out[15]:

topic	bw	publishers	subscribers	n_pub	n_sub

4.3. Clasificación funcional

In [16]:
```
def classify_node(node):
    if any(k in node for k in ["camera", "laser", "points", "image", "dep
        return "perception"
    elif "move_base" in node:
        return "planning"
    elif "amcl" in node:
        return "localization"
    elif "costmap" in node:
        return "mapping"
    elif "cmd_vel" in node:
        return "control"
    else:
        return "system"

df_dep["pub_type"] = df_dep["publishers"].apply(lambda x: [classify_node(
df_dep["sub_type"] = df_dep["subscribers"].apply(lambda x: [classify_node
```

4.4.Clasificación funcional agragada

In [17]:
```
from collections import Counter

def dominant_type(types):
    if not types:
        return "unknown"
    return Counter(types).most_common(1)[0][0]

df_dep["main_type"] = df_dep["pub_type"].apply(dominant_type)
df_dep.groupby("main_type")["bw"].sum().sort_values(ascending=False)
```

Out[17]:
```
main_type
system      96.708680
planning     3.163588
Name: bw, dtype: float64
```

4.5. Conclusiones

El análisis estructural revela que los *topics* de mayor ancho de banda suelen presentar un alto número de suscriptores, amplificando su impacto en la red.

La clasificación funcional muestra que la carga del sistema está dominada por componentes de percepción, mientras que los módulos de planificación y control tienen un impacto significativamente menor en términos de tráfico.

Esto sugiere que el sistema sigue una arquitectura típica de robótica: alta generación de datos en sensores y consumo distribuido en múltiples módulos.

5. Dinámica temporal

5.1. Métricas

In [18]:
```
df_dep["lat_mean_ms"] = df_dep["topic"].apply(lambda t: topics[t].get("la
df_dep["jitter_ms"] = df_dep["topic"].apply(lambda t: topics[t].get("jitt
df_dep["peak_bw"] = df_dep["topic"].apply(lambda t: topics[t].get("peak_b
```

In [19]:
```
df_dep.sort_values(
    by=["lat_mean_ms","bw"],
    ascending=[False, False]
)[["topic","bw","lat_mean_ms"]].head(10)
```

Out[19]:

	topic	bw	lat_mean_ms
3	/robot/robot_front_ptz_camera/image_raw	6.665071	131.855072
2	/robot/front_rgbd_camera/rgb/image_raw	6.665086	108.101449
1	/robot/front_rgbd_camera/depth/image_raw	8.886586	107.681159
0	/robot/front_rgbd_camera/depth/points	71.088803	105.280193
4	/robot/move_base/GlobalPlanner/potential	3.094075	18.733333
7	/robot/front_laser/scan	0.331207	14.233256
8	/robot/move_base/global_costmap/costmap_updates	0.069513	0.489362
5	/gazebo/link_states	1.972124	0.000000
6	/gazebo/model_states	1.099804	0.000000

In [20]:
```
# Trafico explosivo
df_dep["burst_ratio"] = df_dep.apply(
    lambda r: r["peak_bw"] / r["bw"] if r["bw"] > 0 else 0,
    axis=1
)

df_dep.sort_values(by="burst_ratio", ascending=False)[[
    "topic", "bw", "peak_bw", "burst_ratio"
]].head(10)
```

Out[20]:

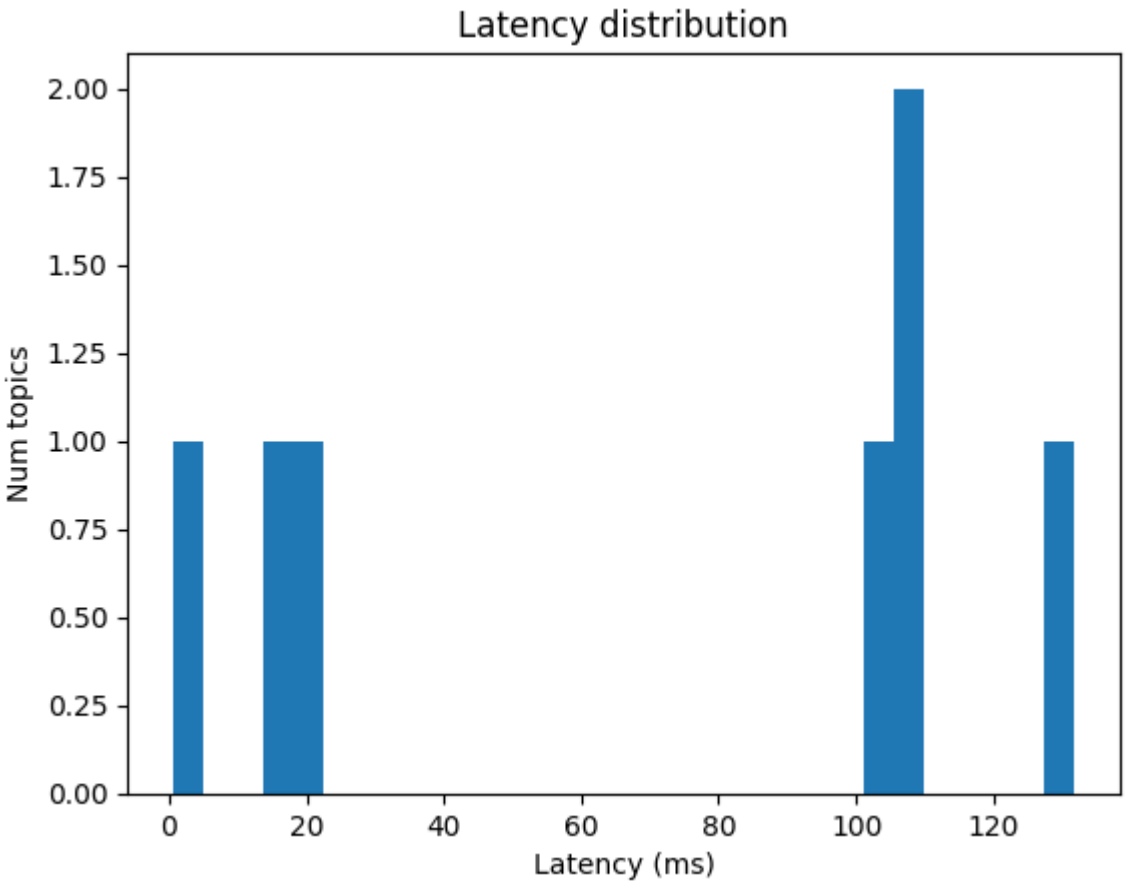
	topic	bw	peak_bw	burst_ratio
7	/robot/front_laser/scan	0.331207	8.317947	25.114075
4	/robot/move_base/GlobalPlanner/potential	3.094075	12.228034	3.952081
2	/robot/front_rgbd_camera/rgb/image_raw	6.665086	13.522867	2.028911
1	/robot/front_rgbd_camera/depth/image_raw	8.886586	16.506423	1.857454
3	/robot/robot_front_ptz_camera/image_raw	6.665071	12.380063	1.857454
0	/robot/front_rgbd_camera/depth/points	71.088803	128.426526	1.806565
8	/robot/move_base/global_costmap/costmap_updates	0.069513	0.080750	1.161646
6	/gazebo/model_states	1.099804	1.103401	1.003271
5	/gazebo/link_states	1.972124	1.977921	1.002939

5.2. Limpiar datos inválidos

In [21]:
```
# no es posible recoger la información de latencia de algunos topics porq
df_dep["lat_available"] = df_dep["lat_mean_ms"] > 0
```

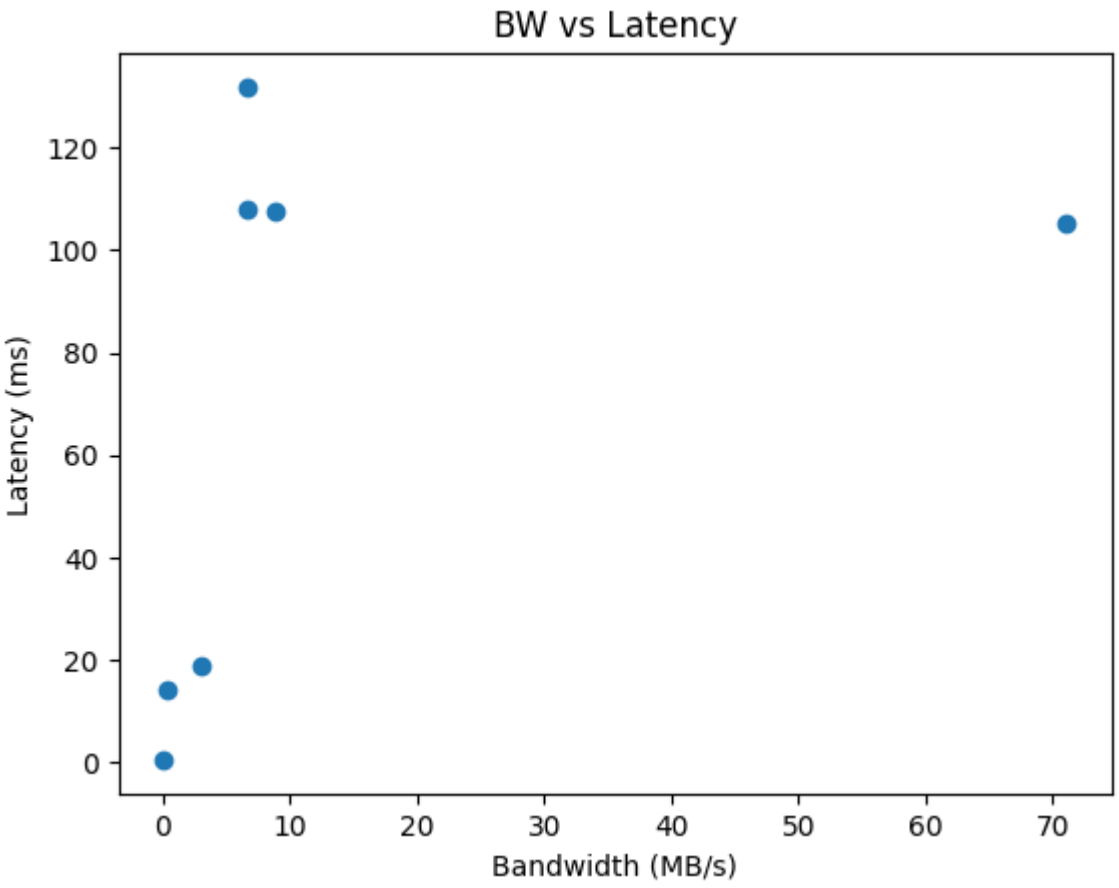
5.3. Distribución de latencia

In [22]:
```
valid = df_dep[df_dep["lat_available"]]
plt.hist(valid["lat_mean_ms"], bins=30)
plt.xlabel("Latency (ms)")
plt.ylabel("Num topics")
plt.title("Latency distribution")
plt.show()
```



5.4. Relación BW vs Latencia

In [23]:
```
valid = df_dep[df_dep["lat_available"]]
plt.scatter(valid["bw"], valid["lat_mean_ms"])
plt.xlabel("Bandwidth (MB/s)")
plt.ylabel("Latency (ms)")
plt.title("BW vs Latency")
plt.show()
```



5.5. Jitter

In [24]:
```
df_dep.sort_values(by="jitter_ms", ascending=False)[["topic","jitter_ms"]]
```

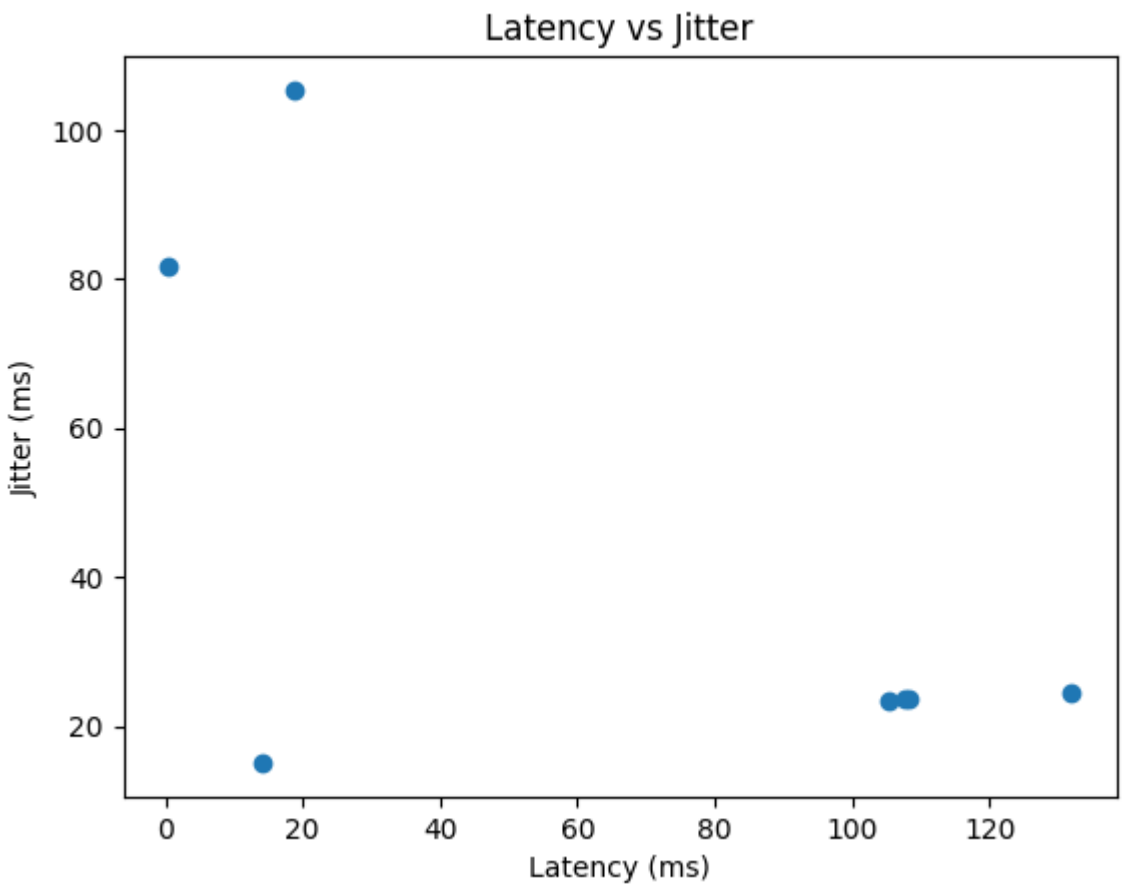
Out[24]:

	topic	jitter_ms
4	/robot/move_base/GlobalPlanner/potential	105.406974
8	/robot/move_base/global_costmap/costmap_updates	81.660682
3	/robot/robot_front_ptz_camera/image_raw	24.367109
1	/robot/front_rgbd_camera/depth/image_raw	23.675922
2	/robot/front_rgbd_camera/rgb/image_raw	23.527657
0	/robot/front_rgbd_camera/depth/points	23.228384
7	/robot/front_laser/scan	15.046288
6	/gazebo/model_states	4.266597
5	/gazebo/link_states	3.714627

5.6. Relación Latencia vs Jitter

In [25]:
```
valid = df_dep[df_dep["lat_available"]]
plt.scatter(valid["lat_mean_ms"], valid["jitter_ms"])
plt.xlabel("Latency (ms)")
plt.ylabel("Jitter (ms)")
plt.title("Latency vs Jitter")
```

```
plt.show()
```



5.7. Correlaciones

In [26]:
```
df_dep[["bw","lat_mean_ms","jitter_ms","n_sub","burst_ratio"]].corr()
```

Out[26]:

	bw	lat_mean_ms	jitter_ms	n_sub	burst_ratio
bw	1.000000	0.454072	-0.133163	-0.235165	-0.173966
lat_mean_ms	0.454072	1.000000	-0.217268	-0.373971	-0.242297
jitter_ms	-0.133163	-0.217268	1.000000	-0.030439	-0.123172
n_sub	-0.235165	-0.373971	-0.030439	1.000000	0.928507
burst_ratio	-0.173966	-0.242297	-0.123172	0.928507	1.000000

In [27]:
```
df_dep[["bw","lat_mean_ms","jitter_ms","n_sub","burst_ratio"]].corr()["la
```

Out[27]:
```
bw             0.454072
lat_mean_ms    1.000000
jitter_ms     -0.217268
n_sub         -0.373971
burst_ratio   -0.242297
Name: lat_mean_ms, dtype: float64
```

5.8. Outliers

In [28]:
```
df_dep[df_dep["lat_mean_ms"] > 500][[
```

```
    "topic","bw","lat_mean_ms","n_sub"
]]
```

Out[28]:

topic	bw	lat_mean_ms	n_sub

In [29]:

```
df_dep[df_dep["burst_ratio"] > 5][[
    "topic","bw","peak_bw","burst_ratio"
]]
```

Out[29]:

	topic	bw	peak_bw	burst_ratio
7	/robot/front_laser/scan	0.331207	8.317947	25.114075

5.9. Cuellos de botella

In [ ]:

```
df_dep[
    (df_dep["lat_mean_ms"] > df_dep["lat_mean_ms"].quantile(0.9)) &
    (df_dep["bw"] > df_dep["bw"].quantile(0.8))
][["topic","bw","lat_mean_ms"]]
```

5.10 Conclusiones

La latencia presenta una distribución heterogénea, con la mayoría de los *topics* en rangos moderados y la presencia de outliers significativos que pueden comprometer el comportamiento del sistema.

No existe una correlación fuerte entre ancho de banda y latencia, lo que indica que los problemas de latencia no dependen únicamente del volumen de datos, sino también de factores como procesamiento, serialización o scheduling.

El jitter muestra variabilidad relevante en ciertos *topics*, lo que puede afectar negativamente a sistemas de control en tiempo real.

El análisis de *burst ratio* evidencia la existencia de tráfico no uniforme en el tiempo, con picos que pueden generar congestión puntual.

En conjunto, el sistema presenta un comportamiento temporal no determinista, con posibles cuellos de botella en *topics* específicos.

6. Impacto final

6.1. Normalización

In [30]:

```
max_bw = df_dep["bw"].max()
valid_lat = df_dep[df_dep["lat_available"]]["lat_mean_ms"]
max_lat = valid_lat.max() if not valid_lat.empty else 0

df_dep["bw_norm"] = df_dep["bw"] / max_bw if max_bw > 0 else 0
df_dep["lat_norm"] = df_dep["lat_mean_ms"] / max_lat if max_lat > 0 else
```

```
df_dep.loc[~df_dep["lat_available"], "lat_norm"] = 0
```

6.2. Clasificación final

In [31]:
```
df_dep["net_impact"] = df_dep["bw"] * df_dep["n_sub"]
df_dep["fanout_ratio"] = df_dep.apply(
    lambda r: r["n_sub"] / r["n_pub"] if r["n_pub"] > 0 else 0,
    axis=1
)
df_dep["fanout_impact"] = df_dep["bw"] * df_dep["fanout_ratio"]
bw_thr = df_dep["bw"].quantile(0.8)
jitter_thr = df_dep["jitter_ms"].quantile(0.9)
def classify(row):
    if not row["lat_available"]:
        return "unknown"
    elif row["bw"] > bw_thr and row["lat_mean_ms"] > 80:
        return "critical"
    elif row["bw"] > bw_thr:
        return "high_bw"
    elif row["jitter_ms"] > jitter_thr:
        return "unstable"
    else:
        return "low_impact"

df_dep["impact_class"] = df_dep.apply(classify, axis=1)
```

6.3. Ranking final

In [32]:
```
df_dep.sort_values(
    by=["impact_class","fanout_impact","net_impact"],
    ascending=[True, False, False]
)[[
    "topic","impact_class","fanout_impact","net_impact","bw","lat_mean_ms
]]
```

Out[32]:

	topic	impact_class	fanout_impact	net_impact	bw	lat_me
0	/robot/front_rgbd_camera/depth/points	critical	71.088803	71.088803	71.088803	105.2
1	/robot/front_rgbd_camera/depth/image_raw	critical	8.886586	8.886586	8.886586	107.6
2	/robot/front_rgbd_camera/rgb/image_raw	low_impact	6.665086	6.665086	6.665086	108.1
3	/robot/robot_front_ptz_camera/image_raw	low_impact	6.665071	6.665071	6.665071	131.8
7	/robot/front_laser/scan	low_impact	1.324826	1.324826	0.331207	14.2
8	/robot/move_base/global_costmap/costmap_updates	low_impact	0.139026	0.139026	0.069513	0.4
5	/gazebo/link_states	unknown	1.972124	1.972124	1.972124	0.0
6	/gazebo/model_states	unknown	1.099804	1.099804	1.099804	0.0
4	/robot/move_base/GlobalPlanner/potential	unstable	3.094075	3.094075	3.094075	18.7

6.4. Descomposición del impacto

In [33]:

```
df_dep["time_impact"] = df_dep["lat_mean_ms"]
df_dep["network_impact"] = df_dep["net_impact"]
df_dep["stability_impact"] = df_dep["jitter_ms"]

df_dep[[
    "topic","network_impact","time_impact","stability_impact"
]].sort_values(by="network_impact", ascending=False)
```

Out[33]:

	topic	network_impact	time_impact	stability_impact
0	/robot/front_rgbd_camera/depth/points	71.088803	105.280193	23.228384
1	/robot/front_rgbd_camera/depth/image_raw	8.886586	107.681159	23.675922
2	/robot/front_rgbd_camera/rgb/image_raw	6.665086	108.101449	23.527657
3	/robot/robot_front_ptz_camera/image_raw	6.665071	131.855072	24.367109
4	/robot/move_base/GlobalPlanner/potential	3.094075	18.733333	105.406974
5	/gazebo/link_states	1.972124	0.000000	3.714627
7	/robot/front_laser/scan	1.324826	14.233256	15.046288
6	/gazebo/model_states	1.099804	0.000000	4.266597
8	/robot/move_base/global_costmap/costmap_updates	0.139026	0.489362	81.660682

In [36]:

```
df_dep.sort_values(
    by=["network_impact","time_impact","stability_impact"],
    ascending=[False, False, False]
)
```

Out[36]:

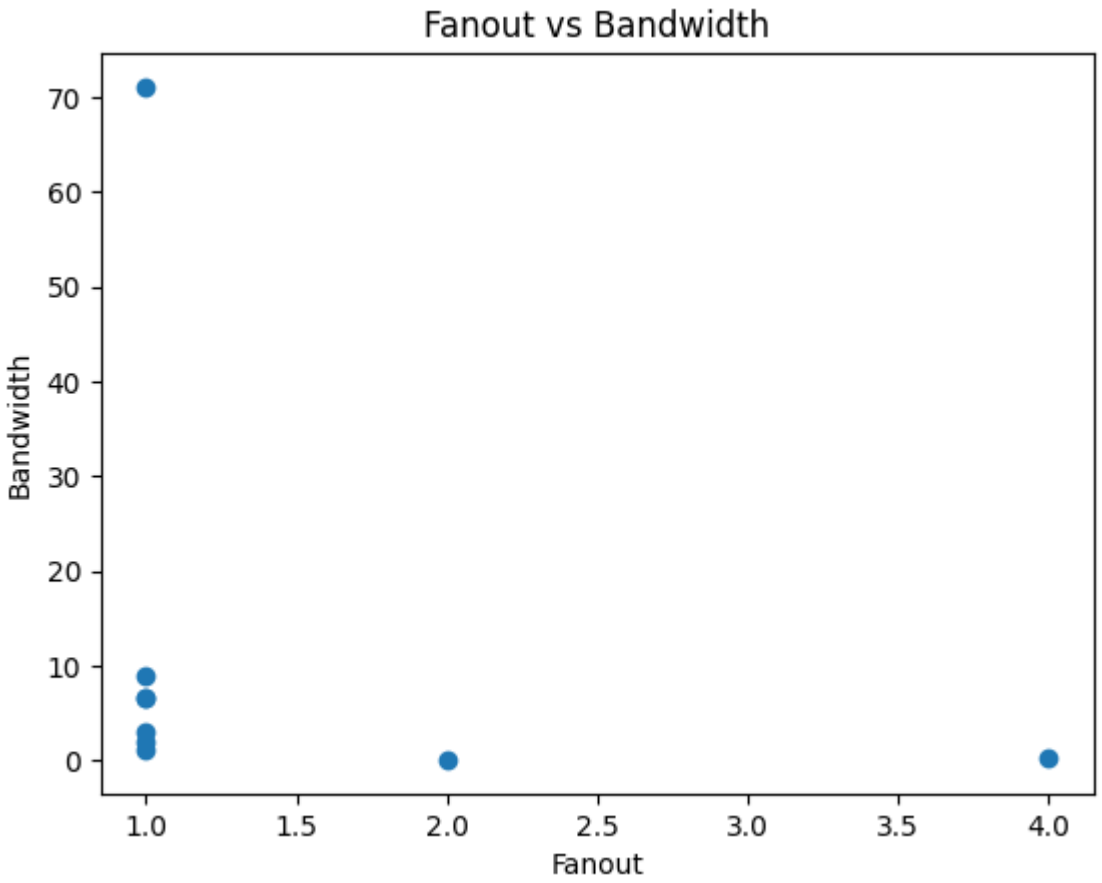
	topic	bw	publishers	subscribers	n_pub	n_sub	pub_t
0	/robot/front_rgbd_camera/depth/points	71.088803	[/gazebo]	[/bag_recorder]	1	1	[syst
1	/robot/front_rgbd_camera/depth/image_raw	8.886586	[/gazebo]	[/bag_recorder]	1	1	[syst
2	/robot/front_rgbd_camera/rgb/image_raw	6.665086	[/gazebo]	[/bag_recorder]	1	1	[syst
3	/robot/robot_front_ptz_camera/image_raw	6.665071	[/gazebo]	[/bag_recorder]	1	1	[syst
4	/robot/move_base/GlobalPlanner/potential	3.094075	[/robot/move_base]	[/bag_recorder]	1	1	[plann
5	/gazebo/link_states	1.972124	[/gazebo]	[/bag_recorder]	1	1	[syst
7	/robot/front_laser/scan	0.331207	[/gazebo]	[/bag_recorder, /robot/amcl, /robot/move_base,...	1	4	[syst
6	/gazebo/model_states	1.099804	[/gazebo]	[/bag_recorder]	1	1	[syst
8	/robot/move_base/global_costmap/costmap_updates	0.069513	[/robot/move_base]	[/bag_recorder, /rviz]	1	2	[plann

9 rows × 23 columns

6.5. Relación fanout vs BW

In [34]:

```
plt.scatter(df_dep["fanout_ratio"], df_dep["bw"])
plt.xlabel("Fanout")
plt.ylabel("Bandwidth")
plt.title("Fanout vs Bandwidth")
plt.show()
```



6.6. Candidatos a offloading

In [37]:
```
df_dep[
    (df_dep["impact_class"] == "critical") &
    (df_dep["fanout_impact"] > df_dep["fanout_impact"].quantile(0.8))
][[
    "topic","bw","lat_mean_ms","fanout_impact"
]]
```

Out[37]:

	topic	bw	lat_mean_ms	fanout_impact
0	/robot/front_rgbd_camera/depth/points	71.088803	105.280193	71.088803
1	/robot/front_rgbd_camera/depth/image_raw	8.886586	107.681159	8.886586

6.7. Conclusiones

El impacto de un *topic* no depende únicamente de su ancho de banda, sino de la combinación de múltiples factores: volumen de datos, latencia y número de suscriptores.

El concepto de *fanout impact* permite identificar *topics* que, aunque no sean los más pesados individualmente, generan una carga significativa al distribuirse a múltiples nodos.

Los *topics* clasificados como críticos combinan alto ancho de banda y alta latencia,

representando los principales cuellos de botella del sistema.

Estos resultados permiten identificar candidatos claros para estrategias de optimización, como compresión, reducción de frecuencia o offloading hacia nodos edge.

7. Conclusiones globales

El sistema analizado presenta una arquitectura fuertemente dominada por flujos de datos de percepción, donde un número reducido de *topics* concentra la mayor parte del tráfico y del impacto en red.

Se ha demostrado que el ancho de banda está principalmente determinado por el tamaño de los mensajes, mientras que la latencia y el jitter dependen de factores adicionales relacionados con procesamiento y distribución.

El análisis estructural y temporal permite identificar *topics* críticos que actúan como cuellos de botella, especialmente aquellos con alto volumen de datos y elevado fanout.

Estos resultados justifican la necesidad de aplicar estrategias de optimización, particularmente mediante offloading hacia arquitecturas edge, donde el procesamiento cercano a la fuente de datos puede reducir tanto la latencia como la carga de red.

En conjunto, el análisis proporciona una base sólida para el diseño de arquitecturas distribuidas eficientes en sistemas robóticos conectados.